

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT	SoA	Fold passes			Map passes		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.3684	0.3268	1.13 ×	0.3049	0.3472	0.88×
Compiler	9	8	0.1276	0.0392	3.26 ×	0.1596	0.1759	0.91×
DBQuery	15	13	0.1254	0.0861	1.46 ×	0.1376	0.1710	0.80×
DecisionTree	7	6	0.5632	0.6483	0.87×	—	—	—
DomTree	15	14	0.1455	0.0612	2.38 ×	0.0596	0.0795	0.75×
KDTree	17	16	0.1817	0.1402	1.30 ×	—	—	—
LinearListReduction	11	11	0.0308	0.0050	6.16 ×	—	—	—
List	2	2	0.1308	0.1123	1.16 ×	0.2514	0.3377	0.74×
MonoTree	3	2	0.0175	0.0225	0.78×	0.0454	0.0430	1.06×
ObjectGraph	5	4	0.0943	0.0815	1.16 ×	0.1429	0.1609	0.89×
PiecewiseFunctions	8	7	0.1399	0.0934	1.50 ×	0.2373	0.2616	0.91×
TernaryTree	5	3	0.0276	0.0268	1.03×	0.0837	0.0790	1.06×
Trie	10	9	0.0642	0.0326	1.97 ×	0.1491	0.1689	0.88×
ColorOctree	25	18	0.0888	0.0761	1.17 ×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
OctTree	1.03×	0.75×	1.45 ×	0.55×	0.26×	0.53×	3.57 ×
Compiler	1.64 ×	0.52×	4.23 ×	0.69×	14.39 ×	0.34×	6.65 ×
DBQuery	1.10 ×	0.47×	1.28 ×	0.29×	8.60 ×	0.23×	6.16 ×
DecisionTree	0.87×	0.69×	0.80×	0.61×	0.44×	0.62×	0.93×
DomTree	1.61 ×	0.57×	3.55 ×	1.47 ×	20.03 ×	0.23×	9.94 ×
KDTree	1.30 ×	0.77×	1.45 ×	0.74×	0.22×	0.75×	4.32 ×
LinearListReduction	6.20 ×	0.90×	34.22 ×	8.61 ×	38.58 ×	8.18 ×	11.72 ×
List	0.82×	0.80×	3.84 ×	0.48×	0.10×	0.87×	1.45 ×
MonoTree	0.97×	0.85×	0.50×	0.76×	0.26×	0.65×	1.22 ×
ObjectGraph	0.98×	0.75×	6.30 ×	0.73×	0.57×	0.72×	2.81 ×
PiecewiseFunctions	1.10 ×	0.60×	2.68 ×	0.98×	2.02 ×	0.81×	4.48 ×
TernaryTree	1.13 ×	0.79×	0.38×	0.47×	0.21×	2.91 ×	1.39 ×
Trie	1.11 ×	0.56×	3.49 ×	0.68×	3.03 ×	0.13×	6.32 ×
Geomean	1.27 ×	0.68×	2.29 ×	0.80×	1.34 ×	0.66×	3.53 ×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
OctTree	2.6e+09/ 2.5e+09	8.2e+09 /1.1e+10	1.5e+07/ 1.1e+07	4.7e+05 /8.6e+05	6.4e+04 /2.5e+05	1.1e+04 /2.0e+04	6.4e+07/ 1.8e+07
Compiler	9.5e+08/ 5.8e+08	9.9e+08 /1.9e+09	1.3e+07/ 3.0e+06	2.2e+05 /3.2e+05	1.2e+06/ 8.5e+04	1.2e+04 /3.6e+04	5.1e+07/ 7.6e+06
DBQuery	9.7e+08/ 8.8e+08	1.1e+09 /2.2e+09	4.0e+06/ 3.1e+06	1.6e+05 /5.4e+05	8.4e+05/ 9.7e+04	6.6e+03 /2.8e+04	2.1e+07/ 3.3e+06
DecisionTree	2.9e+09 /3.3e+09	6.3e+09 /9.0e+09	8.0e+07 /1.0e+08	1.1e+04 /1.8e+04	5.4e+05 /1.2e+06	1.1e+04 /1.7e+04	1.6e+08 /1.7e+08
DomTree	8.6e+08/ 5.4e+08	1.2e+09 /2.2e+09	1.2e+07/ 3.4e+06	1.6e+05/ 1.1e+05	1.0e+06/ 5.1e+04	6.9e+03 /3.0e+04	5.6e+07/ 5.7e+06
KDTree	9.2e+08/ 7.1e+08	3.1e+09 /4.0e+09	1.4e+07/ 9.9e+06	8.1e+02 /1.1e+03	2.4e+04 /1.1e+05	8.2e+02 /1.1e+03	4.4e+07/ 1.0e+07
LinearListReduction	1.6e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.6e+06/ 4.6e+04	3.3e+02/ 3.8e+01	2.7e+05/ 7.0e+03	3.2e+02/ 3.9e+01	1.2e+07/ 1.0e+06
List	1.3e+09 /1.5e+09	4.5e+09 /5.6e+09	9.4e+06/ 2.4e+06	2.9e+05 /6.1e+05	2.0e+04 /2.0e+05	1.3e+04 /1.5e+04	4.8e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.3e+05 /4.6e+05	3.4e+04 /4.4e+04	3.3e+03 /1.3e+04	1.6e+03 /2.5e+03	2.0e+06/ 1.7e+06
ObjectGraph	8.6e+08 /8.8e+08	3.0e+09 /4.0e+09	9.8e+06/ 1.5e+06	1.9e+05 /2.6e+05	3.1e+04 /5.4e+04	1.3e+04 /1.8e+04	2.3e+07/ 8.2e+06
PiecewiseFunctions	1.3e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.4e+07/ 5.3e+06	3.5e+05 /3.6e+05	1.9e+05/ 9.3e+04	2.6e+04 /3.2e+04	4.4e+07/ 9.9e+06
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	5.4e+05 /1.4e+06	8.1e+04 /1.7e+05	4.3e+03 /2.0e+04	2.7e+03/ 9.4e+02	3.3e+06/ 2.4e+06
Trie	6.5e+08/ 5.8e+08	1.2e+09 /2.2e+09	9.0e+06/ 2.6e+06	1.9e+05 /2.7e+05	1.6e+05/ 5.3e+04	6.1e+03 /4.6e+04	2.5e+07/ 4.0e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2872	1.110	0.2151	<i>OOM</i>
DBQuery	0.2629	0.3322	0.2570	1.119
DecisionTree	0.5632	0.9168	0.6483	2.741
DomTree	0.2051	0.3436	0.1407	0.5477
KDTree	0.1817	0.2543	0.1402	0.7016
LinearListReduction	0.0308	0.1332	0.0050	<i>OOM</i>
List	0.3823	<i>OOM</i>	0.4500	<i>OOM</i>
MonoTree	0.0629	0.1407	0.0655	0.2735
ObjectGraph	0.2372	0.4850	0.2423	0.7036
OctTree_sumMass	0.0349	0.0435	0.0273	<i>OOM</i>
OctTree_sumEnergy	0.0358	0.0536	0.0320	<i>OOM</i>
OctTree_countActive	0.0356	0.0449	0.0290	<i>OOM</i>
OctTree_countParticles	0.0355	0.0445	0.0276	<i>OOM</i>
OctTree_barnesHutPotential	0.0415	0.0575	0.0391	<i>OOM</i>
OctTree_fmmPotential	0.0963	0.0965	0.0957	<i>OOM</i>
OctTree_scaleEnergy	0.1525	0.1517	0.1756	<i>OOM</i>
OctTree_clearFlags	0.1524	0.1487	0.1716	<i>OOM</i>
PiecewiseFunctions	0.3771	0.5585	0.3551	2.150
TernaryTree	0.1113	0.1440	0.1058	0.6173
Trie	0.2133	0.2963	0.2016	0.9993
ColorOctree	0.0888	0.1224	0.0761	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.34 $\times$	3.87 $\times$	5.16 $\times$	–
DBQuery	1.02 $\times$	1.26 $\times$	1.29 $\times$	0.30 $\times$
DecisionTree	0.87 $\times$	1.63 $\times$	1.41 $\times$	0.33 $\times$
DomTree	1.46 $\times$	1.68 $\times$	2.44 $\times$	0.63 $\times$
KDTree	1.30 $\times$	1.40 $\times$	1.81 $\times$	0.36 $\times$
LinearListReduction	6.16 $\times$	4.32 $\times$	26.65 $\times$	–
List	0.85 $\times$	–	–	–
MonoTree	0.96 $\times$	2.24 $\times$	2.15 $\times$	0.51 $\times$
ObjectGraph	0.98 $\times$	2.05 $\times$	2.00 $\times$	0.69 $\times$
OctTree_sumMass	1.28 $\times$	1.25 $\times$	1.60 $\times$	–
OctTree_sumEnergy	1.12 $\times$	1.50 $\times$	1.67 $\times$	–
OctTree_countActive	1.23 $\times$	1.26 $\times$	1.55 $\times$	–
OctTree_countParticles	1.29 $\times$	1.26 $\times$	1.61 $\times$	–
OctTree_barnesHutPotential	1.06 $\times$	1.38 $\times$	1.47 $\times$	–
OctTree_fmmPotential	1.01 $\times$	1.00 $\times$	1.01 $\times$	–
OctTree_scaleEnergy	0.87 $\times$	0.99 $\times$	0.86 $\times$	–
OctTree_clearFlags	0.89 $\times$	0.98 $\times$	0.87 $\times$	–
PiecewiseFunctions	1.06 $\times$	1.48 $\times$	1.57 $\times$	0.26 $\times$
TernaryTree	1.05 $\times$	1.29 $\times$	1.36 $\times$	0.23 $\times$
Trie	1.06 $\times$	1.39 $\times$	1.47 $\times$	0.30 $\times$
ColorOctree	1.17 $\times$	1.38 $\times$	1.61 $\times$	–

Table 6: Runtime comparison of Gibbon (AoS/SoA), GHC, and MLton. Times are total median per iteration (s). **Bold** marks the fastest time for each program.

Program	Gibbon-AoS	Gibbon-SoA	GHC	MLton
Compiler	0.2872	0.2151	2.034	–
DBQuery	0.2629	0.2570	2.439	–
DecisionTree	0.5632	0.6483	4.053	–
DomTree	0.2051	0.1407	1.205	–
KDTree	0.1817	0.1402	1.601	–
LinearListReduction	0.0308	0.0050	1.266	–
List	0.3823	0.4500	17.04	–
MonoTree	0.0629	0.0655	1.528	–
ObjectGraph	0.2372	0.2423	2.308	–
OctTree_sumMass	0.0349	0.0273	3.247	–
OctTree_sumEnergy	0.0358	0.0320	5.882	–
OctTree_countActive	0.0356	0.0290	5.257	–
OctTree_countParticles	0.0355	0.0276	5.303	–
OctTree_barnesHutPotential	0.0415	0.0391	5.243	–
OctTree_fmmPotential	0.0963	0.0957	5.783	–
OctTree_scaleEnergy	0.1525	0.1756	6.066	–
OctTree_clearFlags	0.1524	0.1716	5.333	–
PiecewiseFunctions	0.3771	0.3551	3.031	–
TernaryTree	0.1113	0.1058	1.013	–
Trie	0.2133	0.2016	1.728	–
ColorOctree	0.0888	0.0761	7.099	–
Geomean	0.1180	0.1007	3.255	–

Table 7: Gibbon mutable speedups vs GHC. Each entry is total median runtime speedup over all passes for one iteration. *AoS/GHC* and *SoA/GHC* are reported.

Program	AoS/GHC	SoA/GHC
Compiler	0.14 $\times$	0.11 $\times$
DBQuery	0.11 $\times$	0.11 $\times$
DecisionTree	0.14 $\times$	0.16 $\times$
DomTree	0.17 $\times$	0.12 $\times$
KDTree	0.11 $\times$	0.09 $\times$
LinearListReduction	0.02 $\times$	0.00 $\times$
List	0.02 $\times$	0.03 $\times$
MonoTree	0.04 $\times$	0.04 $\times$
ObjectGraph	0.10 $\times$	0.10 $\times$
OctTree_sumMass	0.01 $\times$	0.01 $\times$
OctTree_sumEnergy	0.01 $\times$	0.01 $\times$
OctTree_countActive	0.01 $\times$	0.01 $\times$
OctTree_countParticles	0.01 $\times$	0.01 $\times$
OctTree_barnesHutPotential	0.01 $\times$	0.01 $\times$
OctTree_fmmPotential	0.02 $\times$	0.02 $\times$
OctTree_scaleEnergy	0.03 $\times$	0.03 $\times$
OctTree_clearFlags	0.03 $\times$	0.03 $\times$
PiecewiseFunctions	0.12 $\times$	0.12 $\times$
TernaryTree	0.11 $\times$	0.10 $\times$
Trie	0.12 $\times$	0.12 $\times$
ColorOctree	0.01 $\times$	0.01 $\times$
Geomean	0.04 $\times$	0.03 $\times$

Table 8: Per-pass performance for OctTree, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
barnesHutPotential	F	11/16	31%	0.0415 $\pm$ 0.00084	0.0575 $\pm$ 0.00004	0.0391$\pm$0.00002	–	1.06 $\times$	1.38 $\times$	1.47 $\times$	–
clearFlags	M	15/16	6%	0.1524 $\pm$ 0.00139	0.1487$\pm$0.00110	0.1716 $\pm$ 0.00137	–	0.89 $\times$	0.98 $\times$	0.87 $\times$	–
countActive	F	10/16	38%	0.0356 $\pm$ 0.00002	0.0449 $\pm$ 0.00179	0.0290$\pm$0.00003	–	1.23 $\times$	1.26 $\times$	1.55 $\times$	–
countParticles	F	8/16	50%	0.0355 $\pm$ 0.00082	0.0445 $\pm$ 0.00031	0.0276$\pm$0.00038	–	1.29 $\times$	1.26 $\times$	1.61 $\times$	–
fmmPotential	F	12/16	25%	0.0963 $\pm$ 0.00033	0.0965 $\pm$ 0.00203	0.0957$\pm$0.00220	–	1.01 $\times$	1.00 $\times$	1.01 $\times$	–
scaleEnergy	M	16/16	0%	0.1525 $\pm$ 0.00161	0.1517$\pm$0.00108	0.1756 $\pm$ 0.00024	–	0.87 $\times$	0.99 $\times$	0.86 $\times$	–
sumEnergy	F	12/16	25%	0.0358 $\pm$ 0.00023	0.0536 $\pm$ 0.00177	0.0320$\pm$0.00010	–	1.12 $\times$	1.50 $\times$	1.67 $\times$	–
sumMass	F	10/16	38%	0.0349 $\pm$ 0.00001	0.0435 $\pm$ 0.00184	0.0273$\pm$0.00015	–	1.28 $\times$	1.25 $\times$	1.60 $\times$	–
paletteEntriesQuantized	F	13/25	48%	0.0425 $\pm$ 0.00109	0.0555 $\pm$ 0.00020	0.0333$\pm$0.00004	–	1.28 $\times$	1.31 $\times$	1.67 $\times$	–
quantizationErrorProxy	F	10/25	60%	0.0463 $\pm$ 0.00001	0.0669 $\pm$ 0.00001	0.0428$\pm$0.00001	–	1.08 $\times$	1.44 $\times$	1.56 $\times$	–
Total				0.6733	0.7634	0.6741	–	1.00 $\times$	1.13 $\times$	1.13 $\times$	–
Geomean								1.10 $\times$	1.22 $\times$	1.34 $\times$	–

Table 9: Per-pass PAPI counters for **OctTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.1e+08/ 2.0e+08	8.2e+08 /1.0e+09	7.5e+04 /5.6e+05	2.0e+02/ 6.2e+01	1.9e+03 /2.3e+04	2.0e+02/ 6.3e+01	6.1e+06/ 2.2e+06
clearFlags	M	3.7e+08 /4.2e+08	8.0e+08 /1.7e+09	9.5e+05 /2.4e+06	1.6e+05 /4.4e+05	9.0e+03 /3.7e+04	6.1e+03 /1.0e+04	5.4e+06/ 2.6e+06
countActive	F	1.8e+08/ 1.5e+08	5.8e+08/ 5.8e+08	8.4e+05/ 4.0e+05	1.6e+02/ 4.6e+01	5.3e+03 /1.4e+04	1.7e+02/ 5.0e+01	6.1e+06/ 3.3e+05
countParticles	F	1.8e+08/ 1.4e+08	5.7e+08/ 5.7e+08	9.5e+05/ 2.1e+05	2.4e+02/ 7.1e+01	3.6e+03 /7.6e+03	2.4e+02/ 5.4e+01	6.0e+06/ 6.3e+04
fmmPotential	F	4.9e+08/ 4.8e+08	1.6e+09 /1.8e+09	2.6e+04 /7.2e+05	4.1e+02/ 3.0e+02	1.7e+03 /2.5e+04	4.1e+02/ 3.0e+02	4.4e+06/ 2.2e+06
scaleEnergy	M	3.5e+08 /4.4e+08	9.1e+08 /1.9e+09	1.4e+06 /2.6e+06	3.1e+05 /4.2e+05	1.0e+04 /3.6e+04	2.8e+03 /9.0e+03	5.4e+06/ 2.5e+06
sumEnergy	F	1.8e+08/ 1.6e+08	7.7e+08 /8.9e+08	3.3e+05 /1.1e+06	2.5e+02/ 6.8e+01	2.0e+03 /2.3e+04	2.6e+02/ 6.8e+01	6.3e+06/ 2.6e+06
sumMass	F	1.8e+08/ 1.4e+08	5.7e+08 /6.3e+08	7.3e+05/ 1.9e+05	1.3e+02/ 5.3e+01	2.7e+03 /8.4e+03	1.3e+02/ 5.3e+01	6.1e+06/ 1.2e+06
paletteEntriesQuantized	F	2.2e+08/ 1.7e+08	6.7e+08 /7.6e+08	5.6e+06/ 1.4e+06	2.3e+02/ 1.6e+02	1.1e+04 /4.0e+04	2.3e+02/ 1.5e+02	9.4e+06/ 1.4e+06
quantizationErrorProxy	F	2.3e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.4e+06/ 9.5e+05	6.2e+01 /2.9e+02	1.6e+04 /3.3e+04	6.3e+01 /2.9e+02	9.2e+06/ 3.0e+06
Total		2.6e+09/2.5e+09	8.2e+09/1.1e+10	1.5e+07/1.1e+07	4.7e+05/8.6e+05	6.4e+04/2.5e+05	1.1e+04/2.0e+04	6.4e+07/1.8e+07

Table 10: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
blockCountPass	F	2/9	78%	0.0121 $\pm$ 0.00001	0.0811 $\pm$ 0.00009	0.0019$\pm$0.00000	<i>OOM</i>	6.31$\times$	6.69$\times$	42.27$\times$	–	0.0457	3.78$\times$	23.85
branchStatsPass	F	2/9	78%	0.0118 $\pm$ 0.00001	0.0816 $\pm$ 0.00107	0.0036$\pm$0.00000	<i>OOM</i>	3.31$\times$	6.90$\times$	22.83$\times$	–	0.0409	3.46$\times$	11.45
castInstCountPass	F	3/9	67%	0.0182 $\pm$ 0.00001	0.0816 $\pm$ 0.00118	0.0013$\pm$0.00000	<i>OOM</i>	14.08$\times$	4.49$\times$	63.17$\times$	–	0.0419	2.31$\times$	32.49
has cycle	F	4/9	56%	0.0243 $\pm$ 0.00013	0.0935 $\pm$ 0.00219	0.0172$\pm$0.00082	<i>OOM</i>	1.41$\times$	3.84$\times$	5.43$\times$	–	0.0360	1.48$\times$	2.09
instCountPass	F	2/9	78%	0.0121 $\pm$ 0.00001	0.0813 $\pm$ 0.00007	0.0019$\pm$0.00000	<i>OOM</i>	6.22$\times$	6.71$\times$	41.75$\times$	–	0.6519	53.79$\times$	334.67
latencyModelPass	F	3/9	67%	0.0126 $\pm$ 0.00005	0.0815 $\pm$ 0.00124	0.0030$\pm$0.00000	<i>OOM</i>	4.23$\times$	6.47$\times$	27.37$\times$	–	0.0941	7.47$\times$	31.59
memoryOpStatsPass	F	3/9	67%	0.0119 $\pm$ 0.00001	0.0817 $\pm$ 0.00138	0.0038$\pm$0.00000	<i>OOM</i>	3.11$\times$	6.87$\times$	21.38$\times$	–	0.0423	3.56$\times$	11.07
stripSideEffectsPass	M	7/9	22%	0.0800$\pm$0.00016	0.1883 $\pm$ 0.00165	0.0877 $\pm$ 0.00024	<i>OOM</i>	0.91 $\times$	2.35$\times$	2.15$\times$	–	0.3168	3.96$\times$	3.61
targetReturnPass	M	9/9	0%	0.0796$\pm$0.00105	0.1874 $\pm$ 0.00197	0.0882 $\pm$ 0.00009	<i>OOM</i>	0.90 $\times$	2.35$\times$	2.12$\times$	–	0.2216	2.78$\times$	2.51
throughputModelPass	F	3/9	67%	0.0121 $\pm$ 0.00001	0.0818 $\pm$ 0.00091	0.0030$\pm$0.00000	<i>OOM</i>	4.06$\times$	6.76$\times$	27.43$\times$	–	0.0935	7.73$\times$	31.35
verifyIR	F	9/9	0%	0.0124 $\pm$ 0.00005	0.0705 $\pm$ 0.00144	0.0035$\pm$0.00005	<i>OOM</i>	3.60$\times$	5.67$\times$	20.42$\times$	–	0.4496	36.12$\times$	130.16
Total				0.2872	1.110	0.2151	–	1.34$\times$	3.87$\times$	5.16$\times$	–	2.034	7.08$\times$	9.46
Geomean								3.24$\times$	5.02$\times$	16.24$\times$	–		5.67$\times$	18.37

Table 11: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.1e+07/ 9.7e+06	5.1e+07/ 4.5e+07	1.3e+06/ 6.6e+03	1.9e+02/ 2.7e+01	2.3e+05/ 1.1e+03	1.9e+02/ 2.7e+01	4.7e+06/ 3.4e+03
branchStatsPass	F	6.0e+07/ 1.8e+07	7.9e+07 /8.6e+07	8.3e+05/ 2.1e+04	1.2e+02/ 4.7e+01	5.1e+04/ 3.2e+03	1.3e+02/ 4.9e+01	4.6e+06/ 4.3e+05
castInstCountPass	F	9.2e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.3e+04 /2.4e+04	5.9e+01/ 2.3e+01	7.6e+02 /1.1e+03	6.4e+01/ 2.4e+01	4.7e+06/ 1.3e+03
has cycle	F	1.2e+08/ 8.7e+07	1.1e+08 /1.5e+08	2.3e+06/ 2.9e+05	4.5e+02/ 1.2e+02	1.4e+05/ 2.2e+04	4.4e+02/ 1.2e+02	5.1e+06/ 2.4e+06
instCountPass	F	6.1e+07/ 9.9e+06	5.6e+07 /5.7e+07	1.3e+06/ 1.4e+04	1.4e+02/ 3.0e+01	2.2e+05/ 1.1e+03	1.4e+02/ 3.0e+01	4.7e+06/ 3.8e+03
latencyModelPass	F	6.2e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.4e+04	2.7e+02/ 3.9e+01	2.0e+05/ 3.7e+03	2.7e+02/ 3.9e+01	4.7e+06/ 5.2e+05
memoryOpStatsPass	F	6.0e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.7e+05/ 2.9e+04	1.6e+02/ 4.4e+01	3.2e+04/ 3.1e+03	1.6e+02/ 4.5e+01	4.6e+06/ 4.3e+05
stripSideEffectsPass	M	1.5e+08 /1.9e+08	1.8e+08 /6.0e+08	1.4e+06/ 1.2e+06	1.1e+05 /1.3e+05	6.8e+03 /1.9e+04	4.8e+03 /1.3e+04	4.0e+06/ 1.3e+06
targetReturnPass	M	1.5e+08 /1.9e+08	1.7e+08 /6.3e+08	1.6e+06/ 1.3e+06	1.0e+05 /1.9e+05	6.3e+03 /2.3e+04	5.4e+03 /2.2e+04	4.1e+06/ 1.6e+06
throughputModelPass	F	6.1e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.2e+04	4.1e+02/ 4.5e+01	2.0e+05/ 3.5e+03	4.1e+02/ 4.4e+01	4.7e+06/ 5.4e+05
verifyIR	F	6.3e+07/ 1.7e+07	7.4e+07 /8.6e+07	9.1e+05/ 1.7e+04	2.5e+02/ 1.1e+02	1.4e+05/ 3.9e+03	2.5e+02/ 1.1e+02	4.6e+06/ 4.5e+05
Total		9.5e+08/5.8e+08	9.9e+08/1.9e+09	1.3e+07/3.0e+06	2.2e+05/3.2e+05	1.2e+06/8.5e+04	1.2e+04/3.6e+04	5.1e+07/7.6e+06

Table 12: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/–	1.6e+08/–	2.9e+06/–	5.5e+02/–	1.7e+05/–	5.5e+02/–	1.4e+07/–
branchStatsPass	F	4.1e+08/–	1.8e+08/–	2.7e+06/–	5.2e+02/–	1.3e+05/–	5.1e+02/–	1.4e+07/–
castInstCountPass	F	4.1e+08/–	1.8e+08/–	2.7e+06/–	5.5e+02/–	1.3e+05/–	5.5e+02/–	1.4e+07/–
has cycle	F	4.7e+08/–	2.1e+08/–	2.5e+06/–	3.7e+02/–	6.9e+04/–	3.7e+02/–	1.7e+07/–
instCountPass	F	4.1e+08/–	1.6e+08/–	2.9e+06/–	8.2e+02/–	1.8e+05/–	8.3e+02/–	1.4e+07/–
latencyModelPass	F	4.1e+08/–	1.7e+08/–	2.6e+06/–	6.6e+02/–	1.3e+05/–	6.6e+02/–	1.4e+07/–
memoryOpStatsPass	F	4.1e+08/–	1.9e+08/–	2.6e+06/–	6.6e+02/–	1.2e+05/–	6.6e+02/–	1.4e+07/–
stripSideEffectsPass	M	6.9e+08/–	3.3e+08/–	1.7e+06/–	1.6e+05/–	1.7e+04/–	2.1e+04/–	2.3e+07/–
targetReturnPass	M	6.9e+08/–	3.2e+08/–	1.9e+06/–	1.1e+05/–	1.8e+04/–	2.4e+04/–	2.3e+07/–
throughputModelPass	F	4.1e+08/–	1.7e+08/–	2.7e+06/–	5.3e+02/–	1.3e+05/–	5.3e+02/–	1.4e+07/–
verifyIR	F	3.6e+08/–	1.2e+08/–	2.5e+06/–	2.6e+02/–	1.8e+05/–	2.6e+02/–	1.1e+07/–
Total		5.1e+09/–	2.2e+09/–	2.8e+07/–	2.8e+05/–	1.3e+06/–	4.9e+04/–	1.7e+08/–

Table 13: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
clearQueryFlags	M	14/15	7%	0.0691$\pm$0.00081	0.0763 $\pm$ 0.00009	0.0848 $\pm$ 0.00080	0.2932 $\pm$ 0.00213	0.81 $\times$	1.10$\times$	0.90 $\times$	0.26 $\times$	0.5760	8.34$\times$	
countJoins	F	3/15	80%	0.0195 $\pm$ 0.00001	0.0290 $\pm$ 0.00001	0.0110$\pm$0.00000	0.0864 $\pm$ 0.00004	1.78$\times$	1.49$\times$	2.65$\times$	0.34 $\times$	0.0964	4.95$\times$	
filterSelectivitySkew	F	5/15	67%	0.0196 $\pm$ 0.00000	0.0290 $\pm$ 0.00001	0.0124$\pm$0.00001	0.0875 $\pm$ 0.00002	1.58$\times$	1.48$\times$	2.35$\times$	0.33 $\times$	0.1328	6.78$\times$	
hashJoinPressure	F	5/15	67%	0.0203 $\pm$ 0.00001	0.0332 $\pm$ 0.00095	0.0124$\pm$0.00000	0.0872 $\pm$ 0.00019	1.64$\times$	1.63$\times$	2.68$\times$	0.38 $\times$	0.1044	5.13$\times$	
scaleCosts	M	15/15	0%	0.0685$\pm$0.00011	0.0764 $\pm$ 0.00009	0.0862 $\pm$ 0.00005	0.2984 $\pm$ 0.00162	0.79 $\times$	1.12$\times$	0.89 $\times$	0.26 $\times$	0.7521	10.99$\times$	
sumCost	F	6/15	60%	0.0195 $\pm$ 0.00069	0.0289 $\pm$ 0.00016	0.0141$\pm$0.00024	0.0890 $\pm$ 0.00003	1.38$\times$	1.48$\times$	2.04$\times$	0.32 $\times$	0.5432	27.89$\times$	
sumMemory	F	6/15	60%	0.0197 $\pm$ 0.00001	0.0290 $\pm$ 0.00001	0.0145$\pm$0.00002	0.0888 $\pm$ 0.00003	1.36$\times$	1.47$\times$	2.01$\times$	0.33 $\times$	0.1198	6.08$\times$	
sumRows	F	6/15	60%	0.0268 $\pm$ 0.00001	0.0304 $\pm$ 0.00003	0.0217$\pm$0.00002	0.0885 $\pm$ 0.00117	1.23$\times$	1.14$\times$	1.40$\times$	0.34 $\times$	0.1146	4.28$\times$	
Total				0.2629	0.3322	0.2570	1.119	1.02 $\times$	1.26$\times$	1.29$\times$	0.30 $\times$	2.439	9.28$\times$	
Geomean								1.27$\times$	1.35$\times$	1.72$\times$	0.32 $\times$		7.63$\times$	

Table 14: Per-pass PAPI counters for DBQuery (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	2.7e+05 /1.3e+06	8.0e+04 /3.0e+05	3.6e+04/ 2.4e+04	2.6e+03 /1.2e+04	2.4e+06/ 9.3e+05
countJoins	F	9.9e+07/ 5.5e+07	7.4e+07/ 7.1e+07	7.1e+05/ 2.6e+04	3.1e+02/ 6.9e+01	1.7e+05/ 2.1e+03	3.1e+02/ 6.1e+01	2.7e+06/ 1.9e+03
filterSelectivitySkew	F	9.9e+07/ 6.3e+07	1.0e+08 /1.1e+08	5.2e+05/ 7.4e+04	1.8e+02/ 5.1e+01	1.2e+05/ 6.3e+03	1.9e+02/ 5.7e+01	2.7e+06/ 2.2e+05
hashJoinPressure	F	1.0e+08/ 6.3e+07	8.4e+07 /9.9e+07	7.0e+05/ 9.2e+04	2.6e+02/ 8.3e+01	1.8e+05/ 7.1e+03	2.6e+02/ 8.6e+01	2.7e+06/ 1.2e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.7e+05 /1.3e+06	7.4e+04 /2.4e+05	3.5e+04/ 2.7e+04	2.5e+03 /1.6e+04	2.5e+06/ 1.1e+06
sumCost	F	9.9e+07/ 7.2e+07	9.0e+07 /1.2e+08	6.7e+05/ 8.0e+04	2.4e+02/ 6.7e+01	1.5e+05/ 1.1e+04	2.4e+02/ 7.3e+01	2.7e+06/ 3.3e+05
sumMemory	F	1.0e+08/ 7.3e+07	9.0e+07 /1.2e+08	5.7e+05/ 8.9e+04	3.4e+02/ 6.4e+01	1.2e+05/ 1.2e+04	3.5e+02/ 7.1e+01	2.7e+06/ 3.3e+05
sumRows	F	1.4e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.5e+05/ 1.1e+05	1.4e+02/ 7.5e+01	2.2e+04/ 7.4e+03	1.4e+02/ 8.1e+01	2.3e+06/ 2.9e+05
Total		9.7e+08/8.8e+08	1.1e+09/2.2e+09	4.0e+06/3.1e+06	1.6e+05/5.4e+05	8.4e+05/9.7e+04	6.6e+03/2.8e+04	2.1e+07/3.3e+06

Table 15: Per-pass PAPI counters for DBQuery (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.0e+05 /1.9e+07	6.5e+04 /4.4e+05	1.8e+04 /4.5e+04	1.9e+03 /2.8e+04	2.2e+06/ 9.1e+05
countJoins	F	1.5e+08 /4.4e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.4e+05	1.8e+02 /8.8e+02	1.1e+05/ 1.3e+04	1.9e+02/ 1.3e+02	2.3e+06/ 2.4e+04
filterSelectivitySkew	F	1.5e+08 /4.4e+08	2.0e+08 /7.9e+08	1.5e+06/ 8.8e+05	1.8e+02 /8.3e+02	7.7e+04/ 1.7e+04	1.9e+02 /2.5e+02	2.2e+06/ 1.8e+05
hashJoinPressure	F	1.7e+08 /4.4e+08	1.8e+08 /7.7e+08	1.6e+06/ 6.7e+05	1.6e+02 /5.9e+02	9.4e+04/ 1.6e+04	1.6e+02/ 1.6e+02	2.2e+06/ 1.2e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.7e+05 /2.3e+07	9.8e+04 /4.9e+05	1.9e+04 /5.8e+04	2.1e+03 /4.8e+04	2.2e+06/ 1.2e+06
sumCost	F	1.5e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.2e+06	1.7e+02 /8.1e+02	7.1e+04/ 2.1e+04	1.7e+02 /5.4e+02	2.3e+06/ 3.1e+05
sumMemory	F	1.5e+08 /4.5e+08	1.8e+08 /7.8e+08	1.6e+06/ 1.2e+06	1.2e+02 /7.5e+02	7.8e+04/ 2.0e+04	1.2e+02 /5.0e+02	2.2e+06/ 3.2e+05
sumRows	F	1.5e+08 /4.5e+08	2.3e+08 /8.3e+08	1.5e+06/ 1.2e+06	1.6e+02 /1.1e+03	6.8e+04/ 2.1e+04	1.7e+02 /5.2e+02	2.2e+06/ 3.2e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.1e+07/4.8e+07	1.6e+05/9.3e+05	5.3e+05/2.1e+05	5.0e+03/7.8e+04	1.8e+07/3.4e+06

Table 16: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
classify Batch	F	5/7	29%	0.00$\pm$0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.64 $\times$	1.14$\times$	0.73 $\times$	0.64 $\times$	8.00e-05	5.71$\times$	GH
classify Depth	F	4/7	43%	–	–	–	–	–	–	–	–	1.00e-06	–	–
classify tree	F	5/7	29%	–	–	–	–	–	–	–	–	5.00e-06	–	–
countFeatureUses	F	3/7	57%	0.0641$\pm$0.00002	0.1039 $\pm$ 0.00001	0.0753 $\pm$ 0.00002	0.3076 $\pm$ 0.00118	0.85 $\times$	1.62$\times$	1.38$\times$	0.34 $\times$	0.1814	2.83$\times$	–
countLeaves	F	2/7	71%	0.0612$\pm$0.00001	0.1003 $\pm$ 0.00007	0.0677 $\pm$ 0.00024	0.3060 $\pm$ 0.00118	0.90 $\times$	1.64$\times$	1.48$\times$	0.33 $\times$	0.1944	3.18$\times$	–
countNodes	F	2/7	71%	0.0615$\pm$0.00002	0.1001 $\pm$ 0.00031	0.0677 $\pm$ 0.00002	0.3061 $\pm$ 0.00222	0.91 $\times$	1.63$\times$	1.48$\times$	0.33 $\times$	2.671	43.42$\times$	–
countSmallLeaves	F	3/7	57%	0.0642$\pm$0.00068	0.1002 $\pm$ 0.00001	0.0738 $\pm$ 0.00002	0.2837 $\pm$ 0.00205	0.87 $\times$	1.56$\times$	1.36$\times$	0.35 $\times$	0.1629	2.54$\times$	–
inferenceCost	F	2/7	71%	0.0615$\pm$0.00002	0.1016 $\pm$ 0.00150	0.0684 $\pm$ 0.00072	0.3120 $\pm$ 0.00115	0.90 $\times$	1.65$\times$	1.49$\times$	0.33 $\times$	0.1676	2.73$\times$	–
sumImpurity	F	3/7	57%	0.0631$\pm$0.00063	0.1039 $\pm$ 0.00001	0.0735 $\pm$ 0.00098	0.3138 $\pm$ 0.00188	0.86 $\times$	1.65$\times$	1.41$\times$	0.33 $\times$	0.1628	2.58$\times$	–
sumPathLengths	F	3/7	57%	0.0630$\pm$0.00001	0.1043 $\pm$ 0.00002	0.0779 $\pm$ 0.00001	0.3024 $\pm$ 0.00207	0.81 $\times$	1.66$\times$	1.34$\times$	0.34 $\times$	0.1813	2.88$\times$	–
sumSamples	F	3/7	57%	0.0625$\pm$0.00002	0.1002 $\pm$ 0.00112	0.0757 $\pm$ 0.00075	0.2968 $\pm$ 0.00141	0.83 $\times$	1.60$\times$	1.32$\times$	0.34 $\times$	0.1640	2.62$\times$	–
treeDepth	F	2/7	71%	0.0621$\pm$0.00002	0.1022 $\pm$ 0.00089	0.0683 $\pm$ 0.00001	0.3124 $\pm$ 0.00027	0.91 $\times$	1.65$\times$	1.50$\times$	0.33 $\times$	0.1672	2.69$\times$	–
Total				0.5632	0.9168	0.6483	2.741	0.87 $\times$	1.63$\times$	1.41$\times$	0.33 $\times$	4.053	7.20$\times$	–
Geomean								0.84 $\times$	1.57$\times$	1.32$\times$	0.36 $\times$		3.90$\times$	–

Table 17: Per-pass PAPI counters for DecisionTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	4.7e+04 /8.8e+04	1.7e+05 /2.7e+05	2.7e+01 /1.1e+03	3.8e+01 /6.5e+01	0.0e+00 /2.0e+00	0.0e+00/0.0e+00	1.0e+00 /5.2e+01
classify Depth	F	3.4e+02 /9.9e+02	1.3e+03 /2.5e+03	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00
classify tree	F	3.4e+02 /1.0e+03	1.2e+03 /2.4e+03	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00
countFeatureUses	F	3.2e+08 /3.8e+08	7.4e+08 /1.1e+09	9.1e+06 /1.1e+07	1.5e+03 /1.6e+03	4.6e+04 /1.4e+05	1.5e+03 /1.6e+03	1.8e+07 /2.0e+07
countLeaves	F	3.1e+08 /3.4e+08	6.5e+08 /9.3e+08	8.6e+06 /1.0e+07	1.3e+03 /1.6e+03	8.8e+04 /1.4e+05	1.3e+03 /1.6e+03	1.8e+07 /1.9e+07
countNodes	F	3.1e+08 /3.4e+08	6.5e+08 /9.3e+08	8.6e+06 /1.0e+07	9.2e+02 /9.5e+02	8.2e+04 /1.3e+05	9.2e+02 /9.5e+02	1.8e+07 /1.9e+07
countSmallLeaves	F	3.2e+08 /3.7e+08	7.1e+08 /1.0e+09	8.9e+06 /1.2e+07	1.0e+03 /1.6e+03	3.2e+04 /1.5e+05	1.0e+03 /1.6e+03	1.8e+07 /2.0e+07
inferenceCost	F	3.1e+08 /3.5e+08	6.9e+08 /9.8e+08	8.8e+06 /1.0e+07	1.2e+03 /2.5e+03	8.1e+04 /1.4e+05	1.2e+03 /2.5e+03	1.8e+07 /1.9e+07
sumImpurity	F	3.2e+08 /3.7e+08	6.8e+08 /1.0e+09	8.7e+06 /1.1e+07	9.2e+02 /2.2e+03	5.4e+04 /1.2e+05	9.2e+02 /2.2e+03	1.8e+07 /2.0e+07
sumPathLengths	F	3.2e+08 /4.0e+08	7.8e+08 /1.1e+09	9.5e+06 /1.3e+07	1.2e+03 /2.8e+03	4.1e+04 /1.4e+05	1.2e+03 /2.7e+03	1.8e+07 /2.0e+07
sumSamples	F	3.2e+08 /3.8e+08	6.6e+08 /9.7e+08	8.6e+06 /1.2e+07	1.7e+03 /2.3e+03	5.0e+04 /1.4e+05	1.7e+03 /2.2e+03	1.8e+07 /2.0e+07
treeDepth	F	3.1e+08 /3.5e+08	6.9e+08 /9.8e+08	8.9e+06 /1.0e+07	9.0e+02 /1.9e+03	6.9e+04 /1.4e+05	9.0e+02 /1.9e+03	1.8e+07 /1.9e+07
Total		2.9e+09/3.3e+09	6.3e+09/9.0e+09	8.0e+07/1.0e+08	1.1e+04/1.8e+04	5.4e+05/1.2e+06	1.1e+04/1.7e+04	1.6e+08/1.7e+08

Table 18: Per-pass PAPI counters for DecisionTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	5.8e+04 /9.8e+04	2.0e+05 /2.9e+05	4.6e+02 /1.3e+03	6.3e+01/ 3.3e+01	0.0e+00 /2.0e+00	0.0e+00/0.0e+00	1.4e+01 /4.9e+01
classify Depth	F	6.0e+02 /9.3e+02	1.7e+03 /2.6e+03	0.0e+00 /5.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00
classify tree	F	6.2e+02 /9.5e+02	1.7e+03 /2.6e+03	0.0e+00 /5.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00	0.0e+00/0.0e+00
countFeatureUses	F	5.3e+08 /1.6e+09	1.4e+09 /2.9e+09	5.3e+05 /1.7e+06	5.2e+02 /1.8e+03	5.4e+03 /1.8e+05	5.2e+02 /1.8e+03	1.4e+07/ 1.1e+07
countLeaves	F	5.1e+08 /1.6e+09	1.3e+09 /2.8e+09	5.9e+05 /1.5e+06	2.0e+02 /1.1e+03	6.4e+03 /1.8e+05	2.0e+02 /1.1e+03	1.5e+07/ 1.0e+07
countNodes	F	5.1e+08 /1.6e+09	1.3e+09 /2.9e+09	6.1e+05 /1.5e+06	6.2e+02 /1.1e+03	6.4e+03 /1.8e+05	6.2e+02 /1.1e+03	1.5e+07/ 1.0e+07
countSmallLeaves	F	5.1e+08 /1.4e+09	1.3e+09 /2.8e+09	3.8e+05 /2.1e+06	3.8e+02 /8.0e+02	5.8e+03 /1.8e+05	3.8e+02 /8.1e+02	1.5e+07/ 1.2e+07
inferenceCost	F	5.2e+08 /1.6e+09	1.4e+09 /2.9e+09	6.3e+05 /1.6e+06	2.2e+02 /1.0e+03	6.2e+03 /1.8e+05	2.2e+02 /1.0e+03	1.4e+07/ 1.0e+07
sumImpurity	F	5.3e+08 /1.6e+09	1.4e+09 /2.9e+09	5.7e+05 /1.9e+06	4.6e+02 /2.2e+03	5.3e+03 /1.6e+05	4.6e+02 /2.2e+03	1.4e+07/ 1.1e+07
sumPathLengths	F	5.3e+08 /1.5e+09	1.4e+09 /3.0e+09	3.4e+05 /2.2e+06	3.0e+02 /1.6e+03	5.3e+03 /1.7e+05	3.0e+02 /1.6e+03	1.4e+07/ 1.1e+07
sumSamples	F	5.1e+08 /1.5e+09	1.3e+09 /2.9e+09	4.3e+05 /2.0e+06	3.1e+02 /1.7e+03	5.4e+03 /1.7e+05	3.1e+02 /1.7e+03	1.5e+07/ 1.1e+07
treeDepth	F	5.2e+08 /1.6e+09	1.4e+09 /2.9e+09	6.2e+05 /1.7e+06	6.6e+02 /3.3e+03	6.5e+03 /1.9e+05	6.6e+02 /3.3e+03	1.4e+07/ 1.0e+07
Total		4.6e+09/1.4e+10	1.2e+10/2.6e+10	4.7e+06/1.6e+07	3.7e+03/1.5e+04	5.3e+04/1.6e+06	3.7e+03/1.5e+04	1.3e+08/9.8e+07

Table 19: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
SumArea	F	6/15	60%	0.0373 $\pm$ 0.00017	0.0711 $\pm$ 0.00019	0.0193$\pm$0.00003	0.0885 $\pm$ 0.00018	1.93 $\times$	1.91 $\times$	3.68 $\times$	0.80 $\times$	0.8152	21.87 $\times$	0.8152
computeWidths	M	14/15	7%	0.0302 $\pm$ 0.00004	0.0301$\pm$0.00004	0.0446 $\pm$ 0.00004	0.0902 $\pm$ 0.00112	0.68 $\times$	1.00 $\times$	0.68 $\times$	0.33 $\times$	0.1083	3.59 $\times$	0.1083
count styled	F	3/15	80%	0.0361 $\pm$ 0.00001	0.0708 $\pm$ 0.00002	0.0116$\pm$0.00001	0.0982 $\pm$ 0.00001	3.10 $\times$	1.96 $\times$	6.09 $\times$	0.72 $\times$	0.0515	1.43 $\times$	0.0515
find max Bottom	F	5/15	67%	0.0363 $\pm$ 0.00014	0.0704 $\pm$ 0.00001	0.0186$\pm$0.00004	0.0929 $\pm$ 0.00017	1.95 $\times$	1.94 $\times$	3.79 $\times$	0.76 $\times$	0.0674	1.86 $\times$	0.0674
scaleLayout	M	15/15	0%	0.0294$\pm$0.00002	0.0300 $\pm$ 0.00004	0.0350 $\pm$ 0.00100	0.0879 $\pm$ 0.00007	0.84 $\times$	1.02 $\times$	0.86 $\times$	0.34 $\times$	0.0879	2.99 $\times$	0.0879
sumTextWidth	F	3/15	80%	0.0359 $\pm$ 0.00002	0.0713 $\pm$ 0.00100	0.0117$\pm$0.00001	0.0900 $\pm$ 0.00001	3.07 $\times$	1.99 $\times$	6.10 $\times$	0.79 $\times$	0.0747	2.08 $\times$	0.0747
Total				0.2051	0.3436	0.1407	0.5477	1.46 $\times$	1.68 $\times$	2.44 $\times$	0.63 $\times$	1.205	5.88 $\times$	1.205
Geomean								1.65 $\times$	1.56 $\times$	2.59 $\times$	0.58 $\times$		3.30 $\times$	

Table 20: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.9e+08/ 9.8e+07	2.9e+08 /4.9e+08	2.7e+06/ 6.9e+05	6.9e+02/ 6.2e+01	1.6e+05/ 1.2e+04	6.8e+02/ 6.8e+01	1.3e+07/ 1.9e+06
computeWidths	M	6.4e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	7.0e+04/ 4.3e+04	5.4e+03 /1.2e+04	9.1e+02 /1.5e+04	1.2e+06/ 4.4e+05
count styled	F	1.8e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.2e+06/ 2.6e+04	1.4e+03/ 8.8e+01	2.9e+05/ 5.0e+03	1.4e+03/ 8.8e+01	1.4e+07/ 6.4e+05
find max Bottom	F	1.8e+08/ 9.4e+07	2.5e+08 /4.1e+08	2.5e+06/ 4.4e+05	6.9e+02/ 7.1e+01	1.5e+05/ 9.4e+03	6.9e+02/ 7.3e+01	1.4e+07/ 1.6e+06
scaleLayout	M	6.0e+07 /9.0e+07	8.8e+07 /2.9e+08	1.0e+06/ 7.4e+05	8.3e+04/ 6.4e+04	1.7e+04/ 8.8e+03	1.7e+03 /1.5e+04	1.2e+06/ 4.5e+05
sumTextWidth	F	1.8e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 2.1e+04	1.5e+03/ 8.1e+01	4.1e+05/ 4.2e+03	1.5e+03/ 8.1e+01	1.3e+07/ 6.5e+05
Total		8.6e+08/5.4e+08	1.2e+09/2.2e+09	1.2e+07/3.4e+06	1.6e+05/1.1e+05	1.0e+06/5.1e+04	6.9e+03/3.0e+04	5.6e+07/5.7e+06

Table 21: Per-pass PAPI counters for **DomTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.6e+08 /4.5e+08	4.4e+08 /1.8e+09	8.3e+06/ 2.2e+06	5.0e+02 /8.4e+02	1.4e+04 /3.2e+04	5.0e+02 /8.4e+02	1.0e+07/ 1.3e+06
computeWidths	M	6.7e+07 /3.7e+08	1.5e+08 /8.9e+08	1.4e+06 /3.1e+06	4.7e+04 /3.0e+05	6.4e+03 /1.4e+04	7.6e+02 /2.5e+04	1.1e+06/ 4.4e+05
count styled	F	3.6e+08 /5.0e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.2e+06	1.9e+02 /7.2e+02	3.6e+04 /3.6e+04	1.9e+02 /5.3e+02	1.0e+07/ 4.2e+05
find max Bottom	F	3.6e+08 /4.7e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.4e+06	7.4e+01 /4.8e+02	1.4e+04 /2.8e+04	7.7e+01 /4.8e+02	1.0e+07/ 9.6e+05
scaleLayout	M	6.7e+07 /3.6e+08	1.1e+08 /7.6e+08	9.6e+05 /2.9e+06	3.6e+04 /1.5e+05	3.9e+03 /1.6e+04	1.2e+03 /2.4e+04	1.1e+06/ 4.6e+05
sumTextWidth	F	3.6e+08 /4.6e+08	3.9e+08 /1.7e+09	1.1e+07/ 1.3e+06	4.8e+02 /8.1e+02	8.9e+04/ 3.9e+04	4.8e+02 /5.7e+02	1.0e+07/ 4.1e+05
Total		1.6e+09/2.6e+09	2.0e+09/8.6e+09	4.1e+07/1.2e+07	8.5e+04/4.5e+05	1.6e+05/1.6e+05	3.2e+03/5.2e+04	4.3e+07/4.0e+06

Table 22: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
Find nearest Neighbour	F	13/17	24%	0.0281 $\pm$ 0.00005	0.0410 $\pm$ 0.00004	0.0212 $\pm$ 0.00002	0.1143 $\pm$ 0.00128	1.33 $\times$	1.46 $\times$	1.94 $\times$	0.36 $\times$	5.20e-05	0.36 $\times$	0.36 $\times$
countInRange tight_box	F	13/17	24%	0.0284 $\pm$ 0.00001	0.0416 $\pm$ 0.00001	0.0202 $\pm$ 0.00001	0.1195 $\pm$ 0.00020	1.41 $\times$	1.47 $\times$	2.06 $\times$	0.35 $\times$	6.00e-06	0.35 $\times$	0.35 $\times$
photonMappingBenchmark	F	12/17	29%	0.0429 $\pm$ 0.00001	0.0489 $\pm$ 0.00001	0.0409$\pm$0.00001	0.1282 $\pm$ 0.00100	1.05 $\times$	1.14 $\times$	1.19 $\times$	0.38 $\times$	0.9349	21.87 $\times$	0.9349
pointCloudNeighborhood	F	11/17	35%	0.0253 $\pm$ 0.00014	0.0403 $\pm$ 0.00001	0.0168$\pm$0.00001	0.1113 $\pm$ 0.00006	1.51 $\times$	1.59 $\times$	2.40 $\times$	0.36 $\times$	0.3087	12.08 $\times$	0.3087
sumMassInRange	F	14/17	18%	0.0277 $\pm$ 0.00080	0.0417 $\pm$ 0.00002	0.0203 $\pm$ 0.00001	0.1188 $\pm$ 0.00131	1.36 $\times$	1.51 $\times$	2.05 $\times$	0.35 $\times$	6.00e-06	0.35 $\times$	0.35 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0294 $\pm$ 0.00001	0.0408 $\pm$ 0.00014	0.0208$\pm$0.00080	0.1096 $\pm$ 0.00020	1.41 $\times$	1.39 $\times$	1.96 $\times$	0.37 $\times$	0.3573	12.08 $\times$	0.3573
Total				0.1817	0.2543	0.1402	0.7016	1.30 $\times$	1.40 $\times$	1.81 $\times$	0.36 $\times$	1.601	8.11 $\times$	1.601
Geomean								1.34 $\times$	1.42 $\times$	1.89 $\times$	0.36 $\times$		3.30 $\times$	

Table 23: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.9e+06/ 1.7e+06	8.5e+01 /1.5e+02	3.8e+03 /1.7e+04	8.9e+01 /1.6e+02	7.7e+06/ 1.8e+06
countInRange tight_box	F	1.4e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.1e+06/ 1.7e+06	1.4e+02 /1.7e+02	3.7e+03 /2.0e+04	1.4e+02 /1.7e+02	7.7e+06/ 1.9e+06
photonMappingBenchmark	F	2.2e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.5e+06	2.3e+02 /4.2e+02	5.5e+03 /1.6e+04	2.3e+02 /4.1e+02	5.6e+06/ 1.4e+06
pointCloudNeighborhood	F	1.3e+08/ 8.5e+07	3.3e+08 /4.8e+08	2.6e+06/ 8.7e+05	1.5e+02/ 1.2e+02	3.9e+03 /1.8e+04	1.5e+02/ 1.2e+02	7.7e+06/ 1.6e+06
sumMassInRange	F	1.4e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.5e+06/ 2.3e+06	7.6e+01 /1.5e+02	3.9e+03 /2.4e+04	7.7e+01 /1.5e+02	7.6e+06/ 2.2e+06
twoPointCorrelation bin_8_16	F	1.5e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.2e+06/ 7.7e+05	1.2e+02/ 7.7e+01	2.9e+03 /1.3e+04	1.2e+02/ 7.6e+01	7.6e+06/ 1.3e+06
Total		9.2e+08/7.1e+08	3.1e+09/4.0e+09	1.4e+07/9.9e+06	8.1e+02/1.1e+03	2.4e+04/1.1e+05	8.2e+02/1.1e+03	4.4e+07/1.0e+07

Table 24: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.1e+08 /5.8e+08	4.9e+08 /1.3e+09	4.6e+06 /5.2e+06	2.2e+02 /1.5e+03	5.2e+03 /2.1e+04	2.2e+02 /1.5e+03	5.7e+06/ 1.5e+06
countInRange tight_box	F	2.1e+08 /6.1e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	2.0e+02 /1.7e+03	5.2e+03 /2.2e+04	2.0e+02 /1.7e+03	5.6e+06/ 1.5e+06
photonMappingBenchmark	F	2.5e+08 /6.5e+08	1.1e+09 /2.0e+09	3.0e+06 /5.8e+06	2.1e+02 /2.4e+03	3.4e+03 /2.3e+04	2.1e+02 /2.4e+03	5.1e+06/ 1.5e+06
pointCloudNeighborhood	F	2.0e+08 /5.6e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.7e+06	1.8e+02 /1.1e+03	1.4e+04 /1.9e+04	1.9e+02 /1.1e+03	5.8e+06/ 1.4e+06
sumMassInRange	F	2.1e+08 /6.0e+08	4.7e+08 /1.3e+09	5.1e+06 /6.3e+06	1.3e+02 /1.5e+03	4.7e+03 /2.5e+04	1.4e+02 /1.5e+03	5.6e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.1e+08 /5.6e+08	6.1e+08 /1.4e+09	4.8e+06/ 3.7e+06	2.2e+02 /1.1e+03	5.8e+03 /1.9e+04	2.2e+02 /1.1e+03	5.8e+06/ 1.3e+06
Total		1.3e+09/3.6e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	1.2e+03/9.2e+03	3.8e+04/1.3e+05	1.2e+03/9.2e+03	3.4e+07/8.8e+06

Table 25: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
reduction	F	2/11	82%	0.0308 $\pm$ 0.00001	0.1332 $\pm$ 0.00002	0.0050$\pm$0.00001	<i>OOM</i>	6.16 $\times$	4.32 $\times$	26.65 $\times$	–	1.266	41.10 $\times$	253.26 $\times$
Total				0.0308	0.1332	0.0050	–	6.16 $\times$	4.32 $\times$	26.65 $\times$	–	1.266	41.10 $\times$	253.26 $\times$
Geomean								6.16 $\times$	4.32 $\times$	26.65 $\times$	–		41.10 $\times$	253.26 $\times$

Table 26: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.6e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.6e+06/ 4.6e+04	3.3e+02/ 3.8e+01	2.7e+05/ 7.0e+03	3.2e+02/ 3.9e+01	1.2e+07/ 1.0e+06
Total		1.6e+08/2.5e+07	9.0e+07/1.0e+08	1.6e+06/4.6e+04	3.3e+02/3.8e+01	2.7e+05/7.0e+03	3.2e+02/3.9e+01	1.2e+07/1.0e+06

Table 27: Per-pass PAPI counters for `LinearListReduction` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.8e+08/-	2.5e+08/-	4.0e+06/-	3.2e+02/-	3.9e+05/-	3.2e+02/-	2.7e+07/-
Total		6.8e+08/-	2.5e+08/-	4.0e+06/-	3.2e+02/-	3.9e+05/-	3.2e+02/-	2.7e+07/-

Table 28: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
add1 List	M	2/2	0%	0.2514 $\pm$ 0.00202	OOM	0.3377 $\pm$ 0.00033	OOM	0.74 $\times$	-	-	-	6.326	25.16 $\times$	18.74 $\times$
length List	F	1/2	50%	0.0424 $\pm$ 0.00070	OOM	0.0216 $\pm$ 0.00093	OOM	1.96 $\times$	-	-	-	1.227	28.96 $\times$	56.88 $\times$
sumList	F	2/2	0%	0.0441 $\pm$ 0.00001	OOM	0.0456 $\pm$ 0.00081	OOM	0.97 $\times$	-	-	-	1.169	26.49 $\times$	25.65 $\times$
sumList tail recursive	F	2/2	0%	0.0443 $\pm$ 0.00001	OOM	0.0452 $\pm$ 0.00080	OOM	0.98 $\times$	-	-	-	8.322	187.71 $\times$	184.29 $\times$
Total				0.3823	-	0.4500	-	0.85 $\times$	-	-	-	17.04	44.59 $\times$	37.88 $\times$
Geomean								1.09 $\times$	-	-	-		43.63 $\times$	47.38 $\times$

Table 29: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.6e+08	1.9e+09 /2.8e+09	8.0e+06/ 1.2e+06	2.9e+05 /6.1e+05	1.1e+04 /5.6e+04	1.2e+04 /1.4e+04	9.2e+06/ 7.7e+06
length List	F	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05/ 8.9e+04	1.1e+02/ 7.9e+01	2.9e+03 /1.4e+04	1.1e+02/ 7.9e+01	1.3e+07/ 1.1e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	7.0e+05/ 5.4e+05	9.2e+01/ 6.9e+01	3.2e+03 /6.9e+04	9.3e+01/ 7.2e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.8e+05/ 5.7e+05	1.2e+02 /1.8e+02	3.1e+03 /5.9e+04	1.2e+02 /1.9e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.5e+09	4.5e+09/5.6e+09	9.4e+06/2.4e+06	2.9e+05/6.1e+05	2.0e+04/2.0e+05	1.3e+04/1.5e+04	4.8e+07/3.3e+07

Table 30: Per-pass PAPI counters for `List` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		-/-	-/-	-/-	-/-	-/-	-/-	-/-

Table 31: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
add1Tree	M	3/3	0%	0.0454 $\pm$ 0.00040	0.0604 $\pm$ 0.00045	0.0430 $\pm$ 0.00042	0.1127 $\pm$ 0.00084	1.06 $\times$	1.33 $\times$	1.41 $\times$	0.54 $\times$	0.6992	15.41 $\times$	16.74 $\times$
sumTree	F	3/3	0%	0.0090 $\pm$ 0.00000	0.0401 $\pm$ 0.00001	0.0114 $\pm$ 0.00001	0.0803 $\pm$ 0.00002	0.79 $\times$	4.44 $\times$	3.52 $\times$	0.50 $\times$	0.0903	10.01 $\times$	7.74 $\times$
sumTree TailRec	F	3/3	0%	0.0085 $\pm$ 0.00000	0.0401 $\pm$ 0.00001	0.0111 $\pm$ 0.00002	0.0805 $\pm$ 0.00119	0.77 $\times$	4.71 $\times$	3.62 $\times$	0.50 $\times$	0.7383	86.58 $\times$	66.48 $\times$
Total				0.0629	0.1407	0.0655	0.2735	0.96 $\times$	2.24 $\times$	2.15 $\times$	0.51 $\times$	1.528	24.28 $\times$	23.41 $\times$
Geomean								0.86 $\times$	3.03 $\times$	2.62 $\times$	0.51 $\times$		23.72 $\times$	20.48 $\times$

Table 32: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.1e+05 /4.0e+05	3.3e+04 /4.4e+04	1.8e+03 /5.2e+03	1.3e+03 /2.1e+03	5.0e+05/ 4.4e+05
sumTree	F	4.6e+07 /5.8e+07	2.2e+08 /2.6e+08	7.0e+03 /3.4e+04	1.5e+02 /1.9e+02	7.0e+02 /3.9e+03	1.5e+02 /1.9e+02	7.7e+05/ 6.1e+05
sumTree TailRec	F	4.3e+07 /5.6e+07	1.9e+08 /2.2e+08	6.7e+03 /1.9e+04	1.7e+02 /1.8e+02	7.5e+02 /3.8e+03	1.7e+02 /1.8e+02	7.6e+05/ 6.0e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.3e+05/4.6e+05	3.4e+04/4.4e+04	3.3e+03/1.3e+04	1.6e+03/2.5e+03	2.0e+06/1.7e+06

Table 33: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	1.6e+05 /9.8e+05	3.2e+04 /6.2e+04	1.5e+03 /1.6e+04	1.5e+03 /2.9e+03	4.0e+05/ 3.9e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	7.3e+04 /8.9e+05	2.1e+02 /6.9e+02	1.0e+03 /1.3e+04	2.1e+02 /6.0e+02	3.3e+05 /3.6e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.3e+04 /8.8e+05	4.0e+02/ 9.6e+01	1.1e+03 /1.3e+04	3.9e+02/ 9.7e+01	3.3e+05 /3.6e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	3.0e+05/2.7e+06	3.3e+04/6.3e+04	3.6e+03/4.2e+04	2.1e+03/3.6e+03	1.1e+06/1.1e+06

Table 34: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
countLargeItems	F	3/5	40%	0.0133 $\pm$ 0.00000	0.0448 $\pm$ 0.00001	0.0109 $\pm$ 0.00002	0.0462 $\pm$ 0.00017	1.23 $\times$	3.36 $\times$	4.11 $\times$	0.97 $\times$	0.1116	8.36 $\times$	10.48 $\times$
countMarked	F	3/5	40%	0.0132 $\pm$ 0.00001	0.0450 $\pm$ 0.00001	0.0108 $\pm$ 0.00014	0.0469 $\pm$ 0.00001	1.22 $\times$	3.41 $\times$	4.16 $\times$	0.96 $\times$	0.1091	8.25 $\times$	10.48 $\times$
countSurvivors	F	4/5	20%	0.0140 $\pm$ 0.00010	0.0448 $\pm$ 0.00017	0.0132 $\pm$ 0.00000	0.0467 $\pm$ 0.00001	1.06 $\times$	3.20 $\times$	3.39 $\times$	0.96 $\times$	0.1101	7.84 $\times$	8.48 $\times$
deadBytes	F	4/5	20%	0.0138 $\pm$ 0.00001	0.0444 $\pm$ 0.00001	0.0127 $\pm$ 0.00000	0.0471 $\pm$ 0.00000	1.09 $\times$	3.22 $\times$	3.50 $\times$	0.94 $\times$	0.1104	8.00 $\times$	8.48 $\times$
liveBytes	F	4/5	20%	0.0142 $\pm$ 0.00001	0.0438 $\pm$ 0.00001	0.0127 $\pm$ 0.00000	0.0467 $\pm$ 0.00001	1.12 $\times$	3.09 $\times$	3.44 $\times$	0.94 $\times$	0.1108	7.82 $\times$	8.48 $\times$
sumObjIds	F	3/5	40%	0.0129 $\pm$ 0.00000	0.0436 $\pm$ 0.00001	0.0107 $\pm$ 0.00002	0.0475 $\pm$ 0.00001	1.20 $\times$	3.39 $\times$	4.08 $\times$	0.92 $\times$	0.1099	8.54 $\times$	10.48 $\times$
sweepUnmarked	M	5/5	0%	0.0712 $\pm$ 0.00007	0.0876 $\pm$ 0.00006	0.0802 $\pm$ 0.00091	0.1850 $\pm$ 0.00220	0.89 $\times$	1.23 $\times$	1.09 $\times$	0.47 $\times$	0.4905	6.89 $\times$	6.89 $\times$
totalHeapSize	F	3/5	40%	0.0129 $\pm$ 0.00010	0.0436 $\pm$ 0.00098	0.0104 $\pm$ 0.00068	0.0468 $\pm$ 0.00004	1.23 $\times$	3.39 $\times$	4.17 $\times$	0.93 $\times$	0.5064	39.41 $\times$	48.41 $\times$
touchHotObjects	M	5/5	0%	0.0716 $\pm$ 0.00155	0.0873 $\pm$ 0.00005	0.0807 $\pm$ 0.00002	0.1907 $\pm$ 0.00117	0.89 $\times$	1.22 $\times$	1.08 $\times$	0.46 $\times$	0.6494	9.06 $\times$	8.48 $\times$
Total				0.2372	0.4850	0.2423	0.7036	0.98 $\times$	2.05 $\times$	2.00 $\times$	0.69 $\times$	2.308	9.73 $\times$	9.73 $\times$
Geomean								1.09 $\times$	2.64 $\times$	2.89 $\times$	0.81 $\times$		9.63 $\times$	10.48 $\times$

Table 35: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.8e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.2e+06/ 5.1e+04	2.9e+02/ 1.5e+02	3.5e+03 /3.5e+03	2.9e+02/ 1.5e+02	2.8e+06/ 6.3e+05
countMarked	F	6.7e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.2e+06/ 5.4e+04	2.8e+02/ 1.2e+02	3.7e+03 /3.9e+03	2.8e+02/ 1.2e+02	2.8e+06/ 6.3e+05
countSurvivors	F	7.1e+07/ 6.7e+07	3.1e+08 /3.8e+08	9.9e+05/ 8.3e+04	3.5e+02/ 2.2e+02	2.7e+03 /5.3e+03	3.5e+02/ 2.3e+02	2.7e+06/ 1.2e+06
deadBytes	F	7.0e+07/ 6.4e+07	2.7e+08 /3.5e+08	8.2e+05/ 6.5e+04	2.9e+02/ 1.2e+02	2.7e+03 /5.4e+03	3.0e+02/ 1.2e+02	2.7e+06/ 1.2e+06
liveBytes	F	7.2e+07/ 6.4e+07	2.7e+08 /3.5e+08	7.2e+05/ 6.4e+04	3.0e+02/ 1.9e+02	2.4e+03 /5.4e+03	3.0e+02/ 1.9e+02	2.7e+06/ 1.2e+06
sumObjIds	F	6.5e+07/ 5.4e+07	2.5e+08 /2.7e+08	1.4e+06/ 7.8e+04	3.0e+02/ 1.4e+02	3.4e+03 /4.3e+03	3.0e+02/ 1.4e+02	2.7e+06/ 6.1e+05
sweepUnmarked	M	1.9e+08 /2.3e+08	5.5e+08 /8.8e+08	1.0e+06/ 5.1e+05	7.0e+04 /1.5e+05	4.3e+03 /1.1e+04	4.7e+03 /9.0e+03	2.0e+06/ 1.1e+06
totalHeapSize	F	6.5e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.4e+06/ 1.0e+05	2.1e+02/ 1.1e+02	3.3e+03 /3.5e+03	2.1e+02/ 1.1e+02	2.8e+06/ 6.4e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	9.6e+05/ 5.4e+05	1.2e+05/ 1.1e+05	4.8e+03 /1.1e+04	6.2e+03 /7.8e+03	2.0e+06/ 1.1e+06
Total		8.6e+08/8.8e+08	3.0e+09/4.0e+09	9.8e+06/1.5e+06	1.9e+05/2.6e+05	3.1e+04/5.4e+04	1.3e+04/1.8e+04	2.3e+07/8.2e+06

Table 36: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.3e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.6e+05	2.5e+02/ 1.2e+02	5.0e+03 /7.7e+03	2.5e+02/ 1.2e+02	1.8e+06/ 3.7e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 5.1e+05	1.3e+02 /3.7e+02	5.3e+03 /9.7e+03	1.3e+02 /3.7e+02	1.8e+06/ 3.8e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.8e+05 /4.3e+05	3.4e+02/ 9.5e+01	2.2e+03 /6.6e+03	3.4e+02/ 9.5e+01	1.8e+06/ 6.6e+05
deadBytes	F	2.3e+08 /2.4e+08	4.9e+08 /8.1e+08	3.8e+05 /4.4e+05	1.6e+02 /3.3e+02	3.7e+03 /6.3e+03	1.6e+02 /3.3e+02	1.8e+06/ 6.6e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.5e+05 /4.3e+05	2.2e+02/ 8.1e+01	1.8e+03 /6.2e+03	2.2e+02/ 8.3e+01	1.8e+06/ 6.7e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.5e+05/ 5.2e+05	2.0e+02 /2.6e+02	4.8e+03 /9.4e+03	2.0e+02 /2.6e+02	1.7e+06/ 3.6e+05
sweepUnmarked	M	2.7e+08 /7.6e+08	7.6e+08 /2.1e+09	1.8e+05 /2.5e+06	7.4e+04 /1.4e+05	3.2e+03 /2.0e+04	3.0e+03 /5.3e+03	1.8e+06/ 9.2e+05
totalHeapSize	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.4e+05/ 5.2e+05	1.4e+02 /1.7e+02	4.8e+03 /8.7e+03	1.4e+02 /1.7e+02	1.8e+06/ 3.8e+05
touchHotObjects	M	2.7e+08 /7.9e+08	7.7e+08 /2.2e+09	1.9e+05 /2.0e+06	9.6e+04 /1.7e+05	3.1e+03 /2.1e+04	3.0e+03 /4.1e+03	1.8e+06/ 9.5e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	3.3e+06/7.8e+06	1.7e+05/3.1e+05	3.4e+04/9.5e+04	7.4e+03/1.1e+04	1.6e+07/5.3e+06

Table 37: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across $\sim$ iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am
addConstPW	M	8/8	0%	0.1189$\pm$0.00013	0.1312 $\pm$ 0.00114	0.1305 $\pm$ 0.00115	0.4001 $\pm$ 0.00110	0.91 $\times$	1.10$\times$	1.00 $\times$	0.33 $\times$	0.4106	3.45$\times$
autorefineMaxLevel	F	4/8	50%	0.0208$\pm$0.00001	0.0451 $\pm$ 0.00001	0.0212 $\pm$ 0.00001	0.2353 $\pm$ 0.00118	0.98 $\times$	2.17$\times$	2.13$\times$	0.19 $\times$	0.1313	6.32$\times$
compressMass	F	3/8	62%	0.0197 $\pm$ 0.00000	0.0456 $\pm$ 0.00094	0.0115$\pm$0.00001	0.2344 $\pm$ 0.00084	1.72$\times$	2.31$\times$	3.96$\times$	0.19 $\times$	0.1623	8.23$\times$
diffPW	M	8/8	0%	0.1184$\pm$0.00087	0.1327 $\pm$ 0.00012	0.1311 $\pm$ 0.00011	0.4017 $\pm$ 0.00230	0.90 $\times$	1.12$\times$	1.01 $\times$	0.33 $\times$	0.5916	4.99$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0216 $\pm$ 0.00000	0.0473 $\pm$ 0.00002	0.0187$\pm$0.00097	0.2382 $\pm$ 0.00195	1.16$\times$	2.19$\times$	2.53$\times$	0.20 $\times$	0.0596	2.76$\times$
norm2Estimate	F	5/8	38%	0.0254 $\pm$ 0.00088	0.0547 $\pm$ 0.00044	0.0182$\pm$0.00006	0.1952 $\pm$ 0.00118	1.40$\times$	2.15$\times$	3.01$\times$	0.28 $\times$	0.7337	28.85$\times$
pmapCutHistogram	F	4/8	50%	0.0322 $\pm$ 0.00014	0.0558 $\pm$ 0.00001	0.0124$\pm$0.00001	0.2428 $\pm$ 0.00289	2.59$\times$	1.73$\times$	4.48$\times$	0.23 $\times$	0.7796	24.21$\times$
truncateTolViolations	F	3/8	62%	0.0201 $\pm$ 0.00000	0.0462 $\pm$ 0.00001	0.0114$\pm$0.00001	0.2024 $\pm$ 0.00118	1.76$\times$	2.30$\times$	4.04$\times$	0.23 $\times$	0.1624	8.08$\times$
Total					0.3771	0.5585	0.3551	2.150					
Geomean								1.06 $\times$	1.48$\times$	1.57$\times$	0.26 $\times$	3.031	8.04$\times$
								1.34$\times$	1.81$\times$	2.42$\times$	0.24 $\times$		7.82$\times$

Table 38: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.8e+08 /3.3e+08	5.7e+08 /1.2e+09	1.1e+06 /2.0e+06	1.7e+05 /1.8e+05	1.0e+04 /2.3e+04	1.2e+04 /1.2e+04	4.4e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08 /1.1e+08	2.5e+08 /3.7e+08	2.5e+06/ 1.8e+05	5.8e+02/ 9.3e+01	9.4e+03/ 7.0e+03	5.8e+02/ 9.2e+01	6.0e+06/ 8.6e+05
compressMass	F	1.0e+08/ 5.8e+07	2.5e+08 /2.8e+08	2.3e+06/ 3.0e+04	6.3e+02/ 3.0e+02	2.0e+04/ 4.9e+03	6.4e+02/ 3.1e+02	6.2e+06/ 6.2e+05
diffPW	M	2.8e+08 /3.4e+08	5.9e+08 /1.3e+09	8.7e+05 /2.3e+06	1.8e+05/ 1.7e+05	1.1e+04 /2.6e+04	1.1e+04 /1.9e+04	4.5e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.5e+07	3.3e+08 /4.9e+08	2.5e+06/ 2.8e+05	3.7e+02/ 1.7e+02	1.4e+04/ 1.0e+04	3.7e+02/ 1.7e+02	6.2e+06/ 1.5e+06
norm2Estimate	F	1.3e+08/ 9.2e+07	3.4e+08 /4.4e+08	1.2e+06/ 4.0e+05	2.8e+02/ 1.8e+02	1.5e+04/ 9.4e+03	2.7e+02/ 1.8e+02	6.1e+06/ 1.4e+06
pmapCutHistogram	F	1.6e+08/ 6.3e+07	2.8e+08 /3.7e+08	1.6e+06/ 6.2e+04	1.2e+03/ 3.6e+02	9.2e+04/ 7.5e+03	1.2e+03/ 3.6e+02	4.7e+06/ 1.2e+06
truncateTolViolations	F	1.0e+08/ 5.8e+07	2.5e+08 /2.9e+08	2.2e+06/ 3.1e+04	6.5e+02/ 2.3e+02	1.6e+04/ 4.9e+03	6.5e+02/ 2.4e+02	6.2e+06/ 6.2e+05
Total		1.3e+09/1.1e+09	2.8e+09/4.7e+09	1.4e+07/5.3e+06	3.5e+05/3.6e+05	1.9e+05/9.3e+04	2.6e+04/3.2e+04	4.4e+07/9.9e+06

Table 39: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	3.2e+05 /4.8e+06	1.6e+05 /2.3e+05	5.1e+03 /3.8e+04	7.0e+03 /2.8e+04	4.2e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	3.0e+05 /2.1e+06	8.9e+01 /1.2e+03	1.7e+03 /4.1e+04	9.2e+01 /5.2e+02	4.2e+06/ 6.9e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.3e+06	2.2e+02 /8.8e+02	1.6e+03 /3.0e+04	2.3e+02/ 1.6e+02	4.1e+06/ 3.7e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.4e+09	3.3e+05 /5.1e+06	1.4e+05 /3.2e+05	5.5e+03 /4.5e+04	5.6e+03 /2.2e+04	4.2e+06/ 1.9e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.5e+05 /2.6e+06	3.0e+02 /1.8e+03	1.6e+03 /4.2e+04	3.0e+02 /1.2e+03	4.1e+06/ 1.0e+06
norm2Estimate	F	2.8e+08 /9.9e+08	5.1e+08 /1.3e+09	2.5e+05 /2.5e+06	1.4e+02 /9.8e+02	1.4e+03 /4.1e+04	1.4e+02 /8.2e+02	3.9e+06/ 9.5e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.6e+05 /2.0e+06	1.1e+03/ 5.0e+02	1.0e+04 /3.7e+04	1.1e+03/ 3.4e+02	3.9e+06/ 6.7e+05
truncateTolViolations	F	2.3e+08 /1.0e+09	4.4e+08 /1.3e+09	3.1e+05 /1.3e+06	9.6e+02 /1.1e+03	4.3e+03 /2.7e+04	9.6e+02/ 2.5e+02	4.1e+06/ 3.8e+05
Total		2.2e+09/1.0e+10	4.5e+09/1.4e+10	2.7e+06/2.2e+07	3.0e+05/5.6e+05	3.2e+04/3.0e+05	1.5e+04/5.3e+04	3.3e+07/7.8e+06

Table 40: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); $\pm$ shows standard error of the mean across $\sim$ iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am	GHC/Sm
add 1 tree	M	5/5	0%	0.0837 $\pm$ 0.00025	0.0936 $\pm$ 0.00130	0.0790$\pm$0.00122	0.3429 $\pm$ 0.00210	1.06 $\times$	1.12$\times$	1.18$\times$	0.27 $\times$	0.9265	11.07$\times$	11.73$\times$
sum tree	F	5/5	0%	0.0276 $\pm$ 0.00009	0.0504 $\pm$ 0.00205	0.0268$\pm$0.00010	0.2744 $\pm$ 0.00118	1.03 $\times$	1.83$\times$	1.88$\times$	0.18 $\times$	0.0863	3.13$\times$	3.22$\times$
Total				0.1113	0.1440	0.1058	0.6173	1.05 $\times$	1.29$\times$	1.36$\times$	0.23 $\times$	1.013	9.10$\times$	9.57$\times$
Geomean								1.04 $\times$	1.43$\times$	1.49$\times$	0.22 $\times$		5.89$\times$	6.15$\times$

Table 41: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.6e+05 /1.1e+06	8.1e+04 /1.7e+05	3.4e+03 /1.1e+04	2.7e+03/ 7.4e+02	1.5e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.4e+08	4.6e+08 /5.4e+08	7.5e+04 /3.1e+05	6.2e+01 /1.9e+02	9.1e+02 /9.1e+03	6.2e+01 /2.0e+02	1.8e+06/ 1.3e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	5.4e+05/1.4e+06	8.1e+04/1.7e+05	4.3e+03/2.0e+04	2.7e+03/9.4e+02	3.3e+06/2.4e+06

Table 42: Per-pass PAPI counters for TernaryTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.2e+08 /1.6e+09	9.5e+08 /2.4e+09	2.2e+05 /2.3e+06	7.9e+04 /1.9e+05	3.1e+03 /4.4e+04	4.4e+03 /5.0e+03	1.4e+06/ 9.6e+05
sum tree	F	2.6e+08 /1.4e+09	5.8e+08 /1.0e+09	8.6e+04 /2.3e+06	9.7e+01 /1.1e+03	1.4e+03 /4.0e+04	1.0e+02 /2.8e+02	1.3e+06/ 9.1e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	3.1e+05/4.6e+06	7.9e+04/1.9e+05	4.5e+03/8.4e+04	4.5e+03/5.3e+03	2.8e+06/1.9e+06

Table 43: Per-pass performance for Trie, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors), with GHC speedups (GHC/Am, GHC/Sm). Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si	GHC	GHC/Am
autocompleteTopKProxy	F	5/10	50%	0.0143±0.00000	0.0277±0.00001	0.0087±0.00004	0.1068±0.00001	1.66×	1.93×	3.19×	0.26×	0.1003	6.99×
countLazyNodes	F	4/10	60%	0.0129±0.00000	0.0280±0.00101	0.0067±0.00001	0.1079±0.00129	1.92×	2.17×	4.18×	0.26×	0.0896	6.95×
countTerminals	F	3/10	70%	0.0122±0.00000	0.0262±0.00114	0.0058±0.00007	0.0995±0.00130	2.09×	2.14×	4.49×	0.26×	0.0901	7.37×
decayTrieStats	M	10/10	0%	0.0742±0.00116	0.0798±0.00009	0.0854±0.00009	0.2377±0.00164	0.87×	1.08×	0.94×	0.34×	0.7338	9.89×
resetTraversalState	M	10/10	0%	0.0749±0.00081	0.0813±0.00010	0.0836±0.00175	0.2399±0.00023	0.90×	1.09×	0.97×	0.34×	0.1500	2.00×
sumPrefixFreq	F	3/10	70%	0.0124±0.00002	0.0267±0.00013	0.0057±0.00008	0.1038±0.00004	2.17×	2.15×	4.67×	0.26×	0.4762	38.42×
sumSubtreeHints	F	3/10	70%	0.0123±0.00001	0.0266±0.00001	0.0057±0.00001	0.1037±0.00118	2.16×	2.16×	4.66×	0.26×	0.0875	7.08×
Total				0.2133	0.2963	0.2016	0.9993	1.06×	1.39×	1.47×	0.30×	1.728	8.10×
Geomean								1.58×	1.74×	2.75×	0.28×		7.91×

Table 44: Per-pass PAPI counters for Trie (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.3e+07/ 4.4e+07	1.3e+08 /2.1e+08	1.5e+06/ 5.0e+05	6.8e+01/ 3.9e+01	4.6e+04/ 4.8e+03	7.4e+01/ 3.9e+01	3.7e+06/ 7.2e+05
countLazyNodes	F	6.5e+07/ 3.4e+07	1.3e+08 /1.6e+08	1.3e+06/ 2.6e+04	8.2e+01/ 6.6e+01	1.7e+04/ 3.3e+03	8.8e+01/ 6.5e+01	4.0e+06/ 5.6e+05
countTerminals	F	6.2e+07/ 2.9e+07	1.0e+08 /1.2e+08	1.4e+06/ 1.1e+04	1.1e+02/ 6.4e+01	3.1e+04/ 2.3e+03	1.2e+02/ 6.1e+01	3.9e+06/ 2.1e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	9.3e+05 /1.2e+06	8.7e+04 /1.2e+05	6.4e+03 /2.4e+04	3.2e+03 /2.8e+04	3.1e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.1e+08	2.8e+08 /7.0e+08	9.3e+05/ 7.7e+05	9.9e+04 /1.6e+05	7.3e+03 /1.4e+04	2.4e+03 /1.8e+04	3.0e+06/ 9.0e+05
sumPrefixFreq	F	6.3e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 2.9e+04	8.6e+01/ 5.8e+01	2.3e+04/ 2.6e+03	8.9e+01/ 5.8e+01	3.9e+06/ 2.1e+05
sumSubtreeHints	F	6.2e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 2.4e+04	1.1e+02/ 6.0e+01	3.0e+04/ 2.6e+03	1.1e+02/ 6.0e+01	3.9e+06/ 2.1e+05
Total		6.5e+08/5.8e+08	1.2e+09/2.2e+09	9.0e+06/2.6e+06	1.9e+05/2.7e+05	1.6e+05/5.3e+04	6.1e+03/4.6e+04	2.5e+07/4.0e+06

Table 45: Per-pass PAPI counters for Trie (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.4e+08	2.3e+08 /7.5e+08	1.1e+06 /1.5e+06	6.9e+01 /1.2e+03	3.3e+03 /2.7e+04	7.5e+01 /9.9e+02	2.8e+06/ 4.6e+05
countLazyNodes	F	1.4e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	1.1e+02 /1.0e+03	2.7e+03 /2.2e+04	1.2e+02 /3.3e+02	2.7e+06/ 3.1e+05
countTerminals	F	1.3e+08 /5.1e+08	2.0e+08 /7.3e+08	1.2e+06/ 6.4e+05	9.5e+01 /5.4e+02	1.8e+03 /1.6e+04	9.8e+01 /1.6e+02	2.8e+06/ 1.4e+05
decayTrieStats	M	1.9e+08 /9.9e+08	4.5e+08 /2.1e+09	3.0e+05 /4.5e+06	8.4e+04 /1.5e+05	3.9e+03 /2.5e+04	3.0e+03 /4.1e+04	3.0e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /1.0e+09	3.7e+08 /2.0e+09	3.1e+05 /3.1e+06	8.8e+04 /2.1e+05	4.4e+03 /2.6e+04	2.5e+03 /2.4e+04	2.9e+06/ 9.8e+05
sumPrefixFreq	F	1.4e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	6.8e+01 /4.7e+02	1.5e+03 /1.6e+04	6.9e+01 /1.5e+02	2.8e+06/ 1.5e+05
sumSubtreeHints	F	1.4e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	8.9e+01 /8.0e+02	1.5e+03 /1.6e+04	8.9e+01 /1.5e+02	2.8e+06/ 1.4e+05
Total		1.1e+09/4.6e+09	1.9e+09/7.8e+09	7.9e+06/1.2e+07	1.7e+05/3.7e+05	1.9e+04/1.5e+05	6.0e+03/6.6e+04	2.0e+07/3.4e+06